

*****AWS Site to Site VPN Practical Documents*****

- 1) Create one vpc in Mumbai region {ap-south-1}
Name:- Customer_site_vpc(Left & Local Site)---->Physical DC
CIDR:- 192.168.0.0/16
- 2) Create one vpc in N.Virginia region {us-east-1}
Name:- aws_site_vpc(Right & Remote Site)---->AWS Cloud
CIDR:- 10.0.0.0/16

Steps to Work on

Step 1:- Launch a EC2 in public subnet {ap-south-1} which is left site
{Customer_Site}

Name:- Libre_Swan_SFW_Router_Customer_site {acts as a Linux router}
AMI Name:- Amazon Linux 2023
assign Elastic IP (13.127.94.196)
Disable Source & Destination Checking
Now Login using putty and elastic ip (13.127.94.196)
username:- ec2-user
#sudo su -
#hostnamectl
#yum info libreswan
#yum install -y libreswan

Step 2:- ****Now go to aws right side vpc {us-east-1}**** {Now Maximum work to be done in aws right side vpc}

Create CGW (Customer Gateway using above router settings)
Name:-DC_Customer_Gateway-DC_Router
BGP ASN:- Default which is 65000
Elastic IP of Libre_Swan_SFW_Router_Customer_site(13.127.94.196)

Step 3:- Create VPG (Virtual Private Gateway) in aws right side vpc {us-east-1} using default values

Name:- AWS_VPG(Aws_site_router)

Step 4:- attached newly created vpg to aws site vpc

Step 5:- Create site to site vpn between cgw & vpg {**details to be mentioned very carefully**}

name:- S-2-S vpn between DC & AWS
select the correct VPG & CGW created in previous steps
Routing Option :- Static
Static IP prefixes :- and add both vpc CIDR {192.168.0.0/16 & 10.0.0.0/16}

Step 6:- to enable route propagation in aws site vpc in all 3 route tables
Default_rt_aws_site_vpc, pub_rt & priv_rt

Step 7:- go to your vpn and download the document
make an model :- openswan **Note:- do not make any changes here all defaults values

Step 8:- Again go back to left side DC which is customer dc and run all router configuration command as mentoned in above downloaded document on shown below

putty login to libreswan_router server using elastic ip
customer_site(left_site)

1) Open #nano /etc/sysctl.conf and uncomment the following line to enable IP packet forwarding:

```
net.ipv4.ip_forward = 1
net.ipv4.conf.default.rp_filter = 0
net.ipv4.conf.default.accept_source_route = 0
```

to save this file press ctrl+x and then press y and enter

2) Apply the changes in step 1 by executing the command #sysctl -p

3) Open #nano /etc/ipsec.conf and look for the line below. Ensure that the # in front of the line has been removed, then save and exit the file.

```
include /etc/ipsec.d/*.conf
```

to save this file press ctrl+x and then press y and enter

4) Create a new file at #nano /etc/ipsec.d/aws.conf

```
#####below is a new
```

```
line#####
```

```
conn Tunnel1
    authby=secret
    auto=start
    left=%defaultroute
    leftid=13.127.94.196
    right=3.217.212.66
    type=tunnel
    ikelifetime=8h
    keylife=1h
    phase2alg=aes128-sha1;modp2048    ###not there values aws file is 1024
changed to 2048###
    ike=aes128-sha1;modp2048    #####not there values aws file is 1024
changed to 2048###
#    auth=esp    #####not commented in aws file#####
    keyingtries=%forever
    keyexchange=ike
    leftsubnet=192.168.0.0/16
    rightsubnet=10.0.0.0/16
    dpddelay=10
    dpdtimeout=30
    dpdaction=restart_by_peer
```

5)Create a new file at #nano /etc/ipsec.d/aws.secrets if it doesn't already exist, and append this line to the file (be mindful of the spacing!):

```
13.127.94.196 3.217.212.66: PSK "U0uvHqRUmUUz0zd3FVYkoks003Z.Lh8q"
```

6) Now restart IPsec

```
#systemctl restart ipsec
#systemctl enable --now ipsec
#systemctl status ipsec
```

Now check on your aws site vpn one of the tunnel should be showing Up if yes tunnel successful if no then repeat the steps again

```
*****#####*****
```

Step 7):- Now go to customer site (Left side) make entries in all RT's 10.0.0.0/16--->Traffic should pass through libreswan EC2

Step 8):- Testing public & private connectivity between this two sides by pinging private_ip

Make sure public & private instances in both vpc's enable icmp in there SG's